

Stability, Instability, and Interfacial Dynamics of QED Coherence Domains in Open Aqueous Systems

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Abstract

Building upon the macroscopic Quantum Field Theory (QFT) framework of condensed liquid water, this paper investigates the fine physical mechanisms governing the stability, dissipation, and interfacial boundary dynamics of QED Coherence Domains (CDs). While continuous $U(1)$ gauge symmetry breaking establishes a robust phase-locked ground state shielded by a collective energy gap ($\Delta E_{coh} \approx 2.49$ eV), realistic aqueous systems operate as open, non-equilibrium environments constrained by radiative leakage, ionic screening, and surface interactions. Here, we formulate a quantitative field-theoretical model incorporating spatial order parameter gradients ($\nabla\theta$), Debye-Hückel ionic plasma coupling, and non-linear electro-vibrational infrared (IR) thermal pumping. We analyze the boundary conditions at hydrophilic interfaces, deriving the exact pressure profile, the emergence of topological edge modes, and the critical limits for domain extinction vs. expansion. This provides a rigorous physical foundation for the mechanical resilience, osmotic pressure, and charge separation observed in Exclusion Zone (EZ) water.

Keywords: Quantum Electrodynamics (QED), Coherence Domains, Boundary Dynamics, Radiative Leakage, Ionic Debye Screening, Exclusion Zone Water, Non-Equilibrium Thermodynamics

1. Introduction

In a previous work [1], we established a macroscopic Quantum Electrodynamics (QED) framework for liquid water, demonstrating that its structural cohesion and biological properties are governed by Spontaneous Symmetry Breaking (SSB) of the continuous $U(1)$ gauge symmetry. By modeling the dense aqueous medium as an ensemble of two-level pseudo-spins coupled to

quantized zero-point vacuum fluctuations, we proved that crossing a critical condensation density destabilizes the symmetric vacuum into a non-trivial Mexican Hat potential. This phase transition forces a strict macroscopic phase-quadrature lock between the valence electronic clouds and the trapped radiation vector potential, stabilizing Coherence Domains (CDs) protected by a massive Collective Gap Shift ($\Delta E_{coh} \approx 2.49$ eV per molecule). Furthermore, using a generalized Flory-Huggins lattice field mapping, we demonstrated that the entropic scale invariance of these macroscopic coherent blocks suppresses thermal decoherence at ambient temperature ($T = 300$ K).

However, the ideal ground state derived from variational energy minimization in an infinite closed system represents only a baseline. Real-world aqueous, electrochemical, and biological systems are inherently open, heterogeneous, and bounded. A Coherence Domain in bulk water or near a biological membrane continuously interacts with its environment, giving rise to critical physical phenomena at its boundaries:

1. *Radiative Boundary Leakage and IR Pumping:* Trapped electromagnetic field modes continuously leak across the diffuse domain boundary. To prevent instantaneous thermal quenching, the domain must actively harvest ambient infrared (IR) radiation, operating as a Non-Equilibrium Steady State (NESS) dissipative structure.
2. *Ionic Plasma Screening and Fluctuations:* Mobile ions (H_3O^+ , OH^- , and solvated electrolytes like Na^+ , K^+) generate a Debye screening cloud. Via an Anderson-Higgs mechanism, this plasma acts as a fluctuating conductive mirror that seals the domain, while the thermodynamic expulsion of protons generates a massive electro-mechanical interfacial pressure.
3. *Interfacial Hydrophilic Anchoring:* Solid surfaces act as structural templates that lower the quantum nucleation threshold. They unfold microscopic domains into macroscopic Exclusion Zone (EZ) layers, endowing the interfacial water with topological protection and the anomalous rheology of a “Quantum Gel”.

To bridge the gap between ideal QED field theory and macroscopic interfacial biophysics, this paper formulates a detailed theoretical model of the dynamic stability and boundary electro-mechanics of open Coherence Domains.

The paper is organized as follows:

- **Section 2** formulates the spatial Ginzburg-Landau energy functional to describe the

diffuse domain boundary wall and quantum surface tension.

- **Section 3** analyzes the non-linear dynamical coupling equations, radiative leakage, and IR pumping, deriving the critical size limits for quenching vs. macroscopic stabilization.
- **Section 4** incorporates ionic Debye screening, modeling the Anderson-Higgs mass generation, the fluctuating ionic mirror, and the electro-mechanical balance between Maxwell stress and osmotic pressure.
- **Section 5** investigates hydrophilic surface templates, demonstrating how geometrical shielding and phase-locked rigidity drive the CD–EZ transition.
- **Section 6** applies this framework to physiology, explaining the transient quantum melting during cellular action potentials and the non-dissipative microfluidic transport of protons.
- **Section 7** presents our summary and conclusions.

2. Spatial Ginzburg-Landau Field Theory of Bounded Domains

In standard QED coherence literature [2, 3], Coherence Domains (CDs) are frequently treated as idealized, infinite, or perfectly sealed spherical cavities with sharp step-function boundaries. This oversimplification overlooks a fundamental physical reality: in a continuous fluid medium, a local breaking of gauge symmetry generates a continuous spatial profile for the order parameter $\theta(\mathbf{r})$, establishing a non-zero interfacial energy and an inescapable radiative energy exchange with the surrounding incoherent environment.

In this section, we extend the static, spatially homogeneous variational model to a full continuum Ginzburg-Landau field theory [5]. We demonstrate that the boundary of a CD is not an impenetrable wall, but a dynamic domain wall characterized by a finite surface tension and continuous spatial gradients.

2.1. Physical Meaning of Parameters θ and γ and Spatial Functional

To make the present field-theoretical formulation self-contained, let us first recall the physical origin of the fundamental parameters introduced in the microscopic QED model of liquid water [1]:

- **The Matter Order Parameter $\theta \in [0, \pi]$:** In the microscopic description, each water molecule is treated as an effective two-level quantum system with a ground state $|1\rangle$ and an excited Rydberg state $|2\rangle$ separated by an energy difference $\hbar\omega_0 \approx 12.06$ eV (corresponding to the $2p \rightarrow 5d$ transition). Under macroscopic phase-locking with the quantized electromagnetic field, the electronic state of each molecule is described by the coherent superposition:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|1\rangle + \sin\left(\frac{\theta}{2}\right)e^{i\phi}|2\rangle \quad (1)$$

The angle θ represents the *degree of electronic excitation mixing* across the ensemble:

- $\theta = 0$ corresponds to the purely classical, unassociated fluid where all molecules reside in their electronic ground state $|1\rangle$ without collective phase coherence.
- $\theta = \theta_0 > 0$ represents the broken-symmetry ground state inside the Coherence Domain, where valence electrons oscillate collectively in phase-quadrature with the trapped radiation field.

- **The Dimensionless Coupling Parameter γ :** The parameter γ measures the collective coupling strength between the dense molecular dipole ensemble and the vacuum zero-point radiation field:

$$\gamma \equiv \frac{d_{12}}{\hbar\omega_0} \sqrt{\frac{\hbar\omega_0 n}{2\epsilon_0}} \quad (2)$$

where d_{12} is the transition dipole matrix element and $n = N/V$ is the molecular number density. The threshold $\gamma_c = 1/2$ marks a critical condensation density. For liquid water under standard conditions ($n \approx 3.3 \times 10^{28} \text{ m}^{-3}$), $\gamma \approx 0.87 > 1/2$, which destabilizes the trivial symmetric vacuum $\theta = 0$ into a non-trivial Mexican Hat potential with a stable minimum at $\cos \theta_0 = 1/(4\gamma^2) \approx 0.33$ ($\theta_0 \approx 70^\circ$).

In an inhomogeneous medium, the order parameter becomes a continuous spatial field $\theta(\mathbf{r})$. Deep in the coherent domain bulk, $\theta(\mathbf{r}) \rightarrow \theta_0$, while far outside in the incoherent fluid, $\theta(\mathbf{r}) \rightarrow 0$.

Following standard Ginzburg-Landau field theory [5], the total free energy functional of the bounded system is expressed as:

$$F[\theta(\mathbf{r})] = \int d^3r \left[\frac{1}{2} \rho_{CD} |\nabla \theta(\mathbf{r})|^2 + u(\theta(\mathbf{r})) - u(0) \right] \quad (3)$$

where:

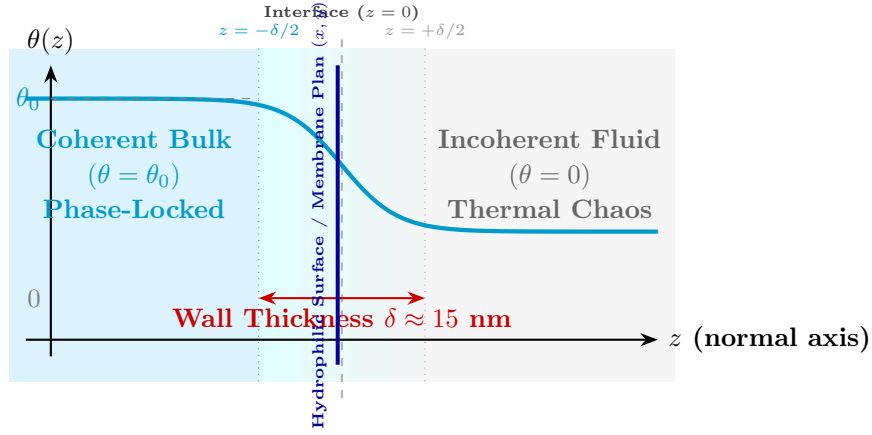


Figure 1: Schematic profile of the order parameter $\theta(z)$ across a Coherence Domain boundary wall. The z -axis is oriented normal to the interface plan (x, y) . The system transitions continuously over a diffuse domain wall of thickness $\delta \approx 15$ nm from the fully phase-locked coherent bulk (θ_0) to the incoherent thermal background ($\theta = 0$).

1. $u(\theta) - u(0) = \hbar\omega_0 n \left(\gamma^2 \cos^2 \theta - \frac{1}{2} \cos \theta - \gamma^2 \right) + \hbar\omega_0 n \left(\frac{1}{2} \right)$ is the local bulk potential energy density. Its minimum at θ_0 provides the collective quantum stabilization energy $\Delta E_{coh} = u(\theta_0) - u(0) \approx -2.49$ eV per molecule.
2. $\frac{1}{2} \rho_{CD} |\nabla \theta|^2$ is the elastic gradient penalty resisting spatial distortions of phase coherence, where $\rho_{CD} \approx \hbar\omega_0 n \lambda_{CD}^2$ represents the quantum phase rigidity ($\lambda_{CD} = c/\omega_0 \approx 100$ nm).

2.2. Domain Wall Profile and Quantum Surface Tension

In classical fluid mechanics, surface tension σ (in J/m² or N/m) represents the excess free energy required to form an interface between two immiscible phases. In a continuous field theory (such as the Ginzburg-Landau or Cahn-Hilliard framework), the interface is not a step discontinuity, but a diffuse transition layer where the order parameter varies smoothly. The excess free energy per unit area stored in this gradient zone constitutes the *Quantum Surface Tension* σ_Q .

Minimizing the functional (3) in 1D along the normal axis z (planar interface approximation, Fig. 1) yields the Euler-Lagrange equation:

$$\rho_{CD} \frac{d^2 \theta(z)}{dz^2} = \frac{\partial u(\theta)}{\partial \theta} \quad (4)$$

Integrating once with the boundary conditions $\theta(-\infty) = \theta_0$ (coherent bulk) and $\theta(+\infty) = 0$ (incoherent fluid) leads to the first-order energy conservation integral:

$$\frac{1}{2}\rho_{CD}\left(\frac{d\theta}{dz}\right)^2 = u(\theta) - u(\theta_0) \quad (5)$$

Solving this differential equation yields a Soliton kink profile for the order parameter [6]:

$$\theta(z) = \frac{\theta_0}{2} \left[1 - \tanh\left(\frac{z}{\delta}\right) \right] \quad (6)$$

where the characteristic domain wall thickness δ is:

$$\delta = \sqrt{\frac{\rho_{CD}}{|u(\theta_0) - u(0)|}} \approx \frac{\lambda_{CD}}{\sqrt{2(\gamma^2 - \frac{1}{4})}} \approx 10 \text{ to } 20 \text{ nm} \quad (7)$$

The *Quantum Surface Tension* σ_Q is obtained by integrating the total excess energy density across the domain wall:

$$\sigma_Q = \int_{-\infty}^{+\infty} \left[\frac{1}{2}\rho_{CD}\left(\frac{d\theta}{dz}\right)^2 + u(\theta(z)) - u(\theta_0) \right] dz = \int_{-\infty}^{+\infty} \rho_{CD}\left(\frac{d\theta}{dz}\right)^2 dz = \sqrt{2\rho_{CD}|u(\theta_0) - u(0)|} \quad (8)$$

For liquid water under standard conditions, Eq. (8) yields $\sigma_Q \approx 10^{-3} \text{ J/m}^2$. This value represents the mechanical interfacial tension generated by the quantum order parameter gradient, which resists domain deformation and imposes a Laplace-like capillary pressure on the boundary.

2.3. Physical Impossibility of Ideal Cavities: Boundary Leakage and Lifetimes

The historical idealization of a Coherence Domain as a “perfectly sealed cavity” with an infinite lifetime is physically untenable in an open fluid. In reality, trapped coherent modes continuously escape across the diffuse boundary wall for two fundamental physical reasons:

1. **Sub-Wavelength Diffuse Boundary and Optical Tunneling:** The local dielectric susceptibility of the medium is directly coupled to the quantum order parameter:

$$\chi(z) = \chi_{inc} + \chi_{CD} \sin^2 \theta(z) \quad (9)$$

It is essential to emphasize that $\chi(z)$ describes the optical response at the extreme ultraviolet (VUV) frequency of the coherent mode ($\hbar\omega_0 \approx 12.06$ eV, $\lambda_{CD} \approx 100$ nm). At this high frequency, far above the standard visible spectrum (where $n_{vis} \approx 1.33$), unexcited and thermally disordered water molecules exhibit negligible collective polarization ($\chi_{inc} \approx 0$). Consequently, the refractive index drops from $n_{bulk} = \sqrt{1 + \chi_{CD} \sin^2 \theta_0} \approx 1.15$ in the coherent core to $n_{out} = \sqrt{1 + \chi_{inc}} \approx 1.00$ in the incoherent outer fluid.

Because this index step occurs across a diffuse domain wall of sub-wavelength thickness ($\delta \approx 15$ nm $\approx \lambda_{CD}/7$), total internal reflection cannot take place. Trapped electromagnetic modes continuously tunnel across the soft interface with a substantial transmission probability per impact: $T_{wall} \approx 0.15$ to 0.30 (see Fig. 2).

2. **Thermal Surface Waves (Ripplons):** At ambient temperature ($T = 300$ K), Brownian collisions from the surrounding fluid deform the soft interface ($\sigma_Q \approx 10^{-3}$ J/m²), exciting capillary surface fluctuations that scatter the trapped coherent field into incoherent radiation.

Kinetic Cavity Model and Intrinsic Passive Lifetimes.. Although the macroscopic coherent field forms a stationary spatial profile ($\langle \mathbf{\Pi} \rangle = \mathbf{0}$), this standing wave physically decomposes into counter-propagating polariton wavepackets. Each progressive component transports energy at the slow polariton group velocity:

$$v_g = \left. \frac{d\omega(k)}{dk} \right|_{k_{CD}} \approx 10^4 \text{ m/s} \ll c \quad (10)$$

where the massive deceleration arises from the strong hybridization between photons and the molecular electronic transitions.

Following kinetic cavity theory (the Cauchy-Crofton mean chord theorem), the average distance traversed by an internal wavepacket between two boundary impacts is $\bar{L} \approx V/S$. The effective collision frequency of the trapped energy with the domain wall is therefore:

$$\nu_{bounce} \approx \left(\frac{v_g}{n_{bulk}} \right) \left(\frac{S}{V} \right) \quad (11)$$

In the absence of an external energy replenishment mechanism ($I_{IR} = 0$), each collision transmits a fraction T_{wall} into the outer medium. The domain leaks its trapped field at an intrinsic dissipation rate $\kappa = 1/\tau = \nu_{bounce} T_{wall}$, yielding the passive lifetime:

$$\tau = \frac{1}{\kappa} = \frac{n_{bulk}}{v_g T_{wall}} \left(\frac{V}{S} \right) \quad (12)$$

Evaluating this expression for representative geometries reveals two critical timescales (see Fig. 3):

- **Spherical Bulk Domains** ($R_{CD} \approx 100$ nm, $S/V = 3/R_{CD}$):

$$\tau_{CD} = \frac{n_{bulk} R_{CD}}{3v_g T_{wall}} \approx 10 \text{ to } 30 \text{ ps} \quad (13)$$

An unpumped spherical CD in bulk water quenches into thermal chaos almost instantaneously.

- **Planar Exclusion Zone (EZ) Layers** ($L_{EZ} \approx 100$ μm , $S/V = 1/L_{EZ}$):

$$\tau_{EZ} = \frac{n_{bulk} L_{EZ}}{v_g T_{wall}} \approx 30 \text{ ns} \quad (14)$$

The planar geometry reduces the surface-to-volume ratio by a factor of 3000, extending the lifetime by three orders of magnitude.

This quantitative result proves that *no Coherence Domain can persist as an isolated, closed equilibrium state*. To achieve macroscopic stability, a CD must operate as an open Non-Equilibrium Steady State (NESS) driven by continuous environmental infrared pumping.

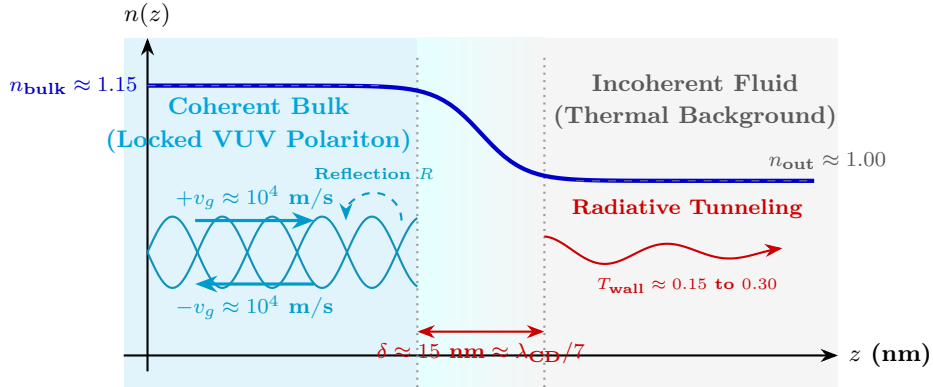


Figure 2: Refractive index step $n(z)$ at $\hbar\omega_0 \approx 12.06$ eV and optical tunneling across the sub-wavelength diffuse boundary ($\delta \approx 15$ nm). Counter-propagating polaritons moving at group velocity $v_g \approx 10^4$ m/s strike the interface and leak out into the incoherent medium ($T_{wall} \approx 0.2$).

3. Non-Linear IR Pumping, Stability Equations, and Critical Bounds

To counteract boundary dissipation, open Coherence Domains harvest ambient infrared (IR) thermal fluctuations. In this section, we formulate the coupled non-linear dynamic rate equa-

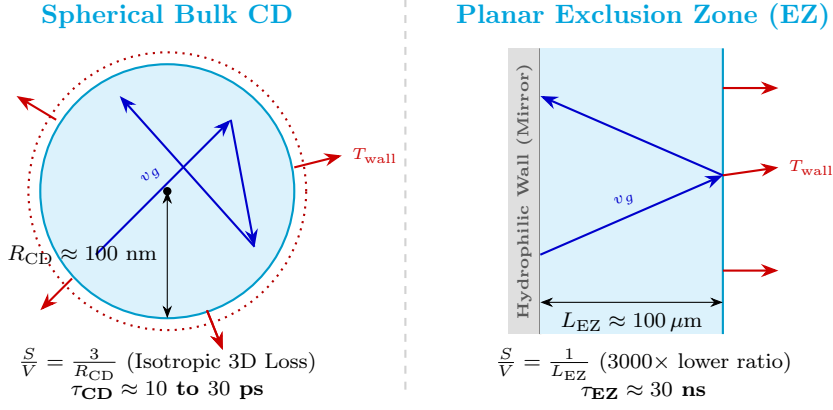


Figure 3: Kinetic bouncing model and geometric scaling of radiative lifetimes: (Left) 3D bulk spherical CD with isotropic boundary leakage and picosecond quenching ($\tau_{\text{CD}} \sim 10$ ps); (Right) Planar EZ layer anchored to a solid template, reducing boundary losses by a factor of 3000 ($\tau_{\text{EZ}} \sim 30$ ns).

tions governing the open system, analyze the resulting steady-state attractor, and define the qualitative geometric nature of the leakage parameter κ .

3.1. Coupled Non-Linear Rate Equations and Stability Analysis

The open electro-vibrational dynamics of a Coherence Domain are described by two coupled dimensionless state variables:

1. $x(t) \equiv \sqrt{\langle \hat{a}^\dagger \hat{a} \rangle} / N$: the normalized amplitude of the trapped coherent UV mode ($\hbar\omega_0 \approx 12.06$ eV). The square $x^2(t)$ represents the average number of coherent photons per water molecule.
2. $y(t) \equiv \langle \hat{b}^\dagger \hat{b} \rangle / N$: the excitation density stored in the intramolecular vibrational reservoir ($O-H$ stretching/bending modes around $\lambda_{\text{IR}} \approx 3 \mu\text{m}$, $\hbar\omega_{\text{IR}} \approx 0.4$ eV).

The fundamental dynamical equations governing the domain are:

$$\frac{dx}{dt} = (gy - \kappa)x \quad (15)$$

$$\frac{dy}{dt} = \sigma_{\text{IR}} I_{\text{IR}} - gyx^2 \quad (16)$$

Physical Justification of Terms in Eqs. (15)–(16):.

- $+\sigma_{IR}I_{IR}$ (**Thermal IR Harvesting**): Represents the continuous absorption of ambient thermal infrared irradiance I_{IR} by the hydrogen-bond vibrational network, where σ_{IR} is the effective molecular cross-section.
- $+gyx$ and $-gyx^2$ (**Non-Linear Optomechanical Pumping**): Represents the non-linear parametric up-conversion transferring energy from the vibrational reservoir (y) into the coherent UV mode (x), with coupling constant g . The rate of field amplitude growth is proportional to yx , which corresponds to an energy depletion rate $-gyx^2$ in the vibrational reservoir. (A formal derivation from the multimode Maxwell-Bloch field theory is detailed in Appendix Appendix A).
- $-\kappa x$ (**Radiative Boundary Loss**): Represents the continuous loss of coherent field amplitude through the diffuse domain boundary at an effective decay rate κ .

Non-Equilibrium Steady State (NESS) and Thermodynamic Efficiency:. Setting $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} = 0$ in Eqs. (15)–(16) determines the non-trivial NESS attractor (x_0, y_0) (see Fig. 4):

$$y_0 = \frac{\kappa}{g}, \quad x_0^2 = \frac{\sigma_{IR}I_{IR}}{\kappa} \quad (17)$$

Equation (17) reveals two essential stability properties:

- The vibrational reservoir clamps at $y_0 = \kappa/g$, directly compensating for boundary leakage.
- The steady-state coherent photon density x_0^2 is directly proportional to the ambient thermal IR pump (I_{IR}) and inversely proportional to the dissipation rate (κ).

The thermodynamic conversion efficiency η_{conv} , defined as the ratio of replaced coherent power loss to absorbed thermal IR power, simplifies at equilibrium to the quantum frequency ratio:

$$\eta_{conv} = \frac{\hbar\omega_0 \cdot N \cdot (gy_0x_0^2)}{\hbar\omega_{IR} \cdot N \cdot (\sigma_{IR}I_{IR})} = \frac{\omega_0}{\omega_{IR}} \approx 30 \quad (18)$$

This confirms that parametric conversion of ambient heat sustains the high-frequency coherent condensate with high efficiency under room-temperature conditions ($T = 300$ K).

Threshold for Macroscopic Stability:. To maintain continuous $U(1)$ gauge symmetry breaking, the steady-state photon density must exceed the critical threshold x_c^2 required to sustain the Mexican Hat potential ($x_0^2 \geq x_c^2$) [1]. From Eq. (17), this imposes an upper bound on the allowable dissipation rate:

$$\kappa \leq \kappa_c \equiv \frac{\sigma_{IR} I_{IR}}{x_c^2} \quad (19)$$

If $\kappa > \kappa_c$, dissipation exceeds the maximum IR pumping capacity; $x(t)$ decays towards zero and the coherent domain collapses.

3.2. Geometric Scaling of κ and Critical Size Limits

We now examine the physical parameters governing the dissipation rate κ and discuss its geometric dependence.

Physical Structure of the Loss Rate κ :. The radiative loss rate κ is governed by the frequency at which internal polariton modes collide with the boundary wall and the probability of escaping per bounce:

$$\kappa \approx \left(\frac{v_g}{n_{bulk}} \right) \left(\frac{S}{V} \right) T_{wall} \quad (20)$$

where $v_g \approx 10^4$ m/s is the polariton group velocity, $n_{bulk} \approx 1.15$ is the refractive index, $T_{wall} \approx 0.2$ is the wall transmission coefficient, and S/V is the surface-to-volume ratio.

Defining the characteristic *boundary leakage velocity* as $v_{leak} \equiv 3v_g T_{wall}/n_{bulk} \approx 10^4$ m/s, the loss rate scales with domain geometry:

- **Spherical Bulk CD (Radius R_{CD} , $S/V = 3/R_{CD}$):**

$$\kappa_{sphere}(R_{CD}) = \frac{v_{leak}}{R_{CD}} \quad (21)$$

- **Planar Interfacial Layer (EZ Thickness L_{EZ} , $S/V = 1/L_{EZ}$):**

$$\kappa_{planar}(L_{EZ}) = \frac{v_{leak}}{3L_{EZ}} \quad (22)$$

The Critical Quenching Radius R_{min} (see Fig. 5):. Substituting $\kappa_{sphere}(R_{CD})$ into the stability criterion (19) yields the coherent field intensity as a function of domain radius:

$$x_0^2(R_{CD}) = \frac{\sigma_{IR} I_{IR}}{v_{leak}} R_{CD} \quad (23)$$

Setting $x_0^2(R_{min}) = x_c^2$ defines the *Critical Quenching Radius* R_{min} :

$$R_{min} = \frac{v_{leak} x_c^2}{\sigma_{IR} I_{IR}} \quad (24)$$

Equation (24) delineates two distinct physical regimes in open water:

1. **Sub-Critical Regime ($R_{CD} < R_{min}$): Transient Quenching.** For embryonic water clusters, the large surface-to-volume ratio produces radiative losses that overwhelm thermal IR harvesting ($\kappa > \kappa_c$). The field drops below threshold ($x < x_c$) and the domain collapses on a picosecond timescale.
2. **Super-Critical Regime ($R_{CD} \geq R_{min}$): Macroscopic NESS Stabilization.** When an external energy fluctuation or an interfacial substrate (Sections 4 and 5) enables a domain to reach R_{min} , thermal IR pumping over-compensates radiative leakage. The domain locks into a stable, self-sustaining Non-Equilibrium Steady State.

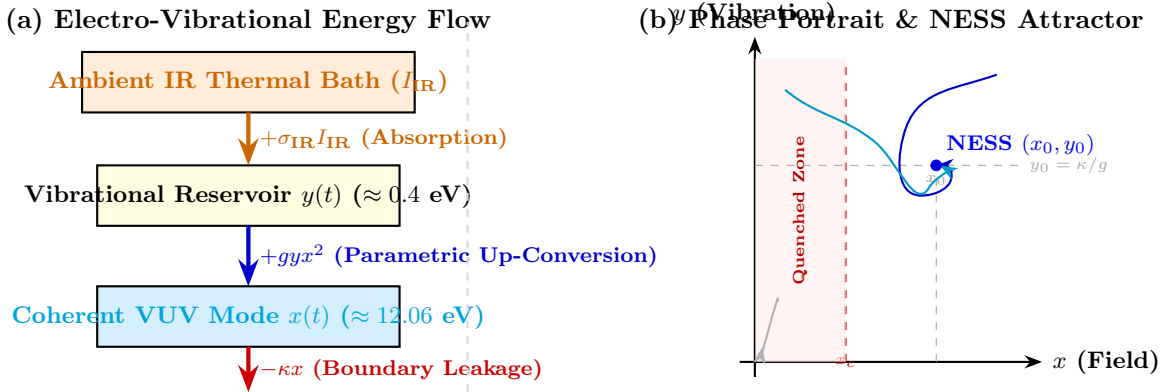


Figure 4: (a) Schematic of non-linear electro-vibrational IR pumping compensating for boundary leakage; (b) Phase-space trajectory (x, y) converging toward the stable Non-Equilibrium Steady State (NESS) attractor (x_0, y_0) above the critical coherence threshold x_c .

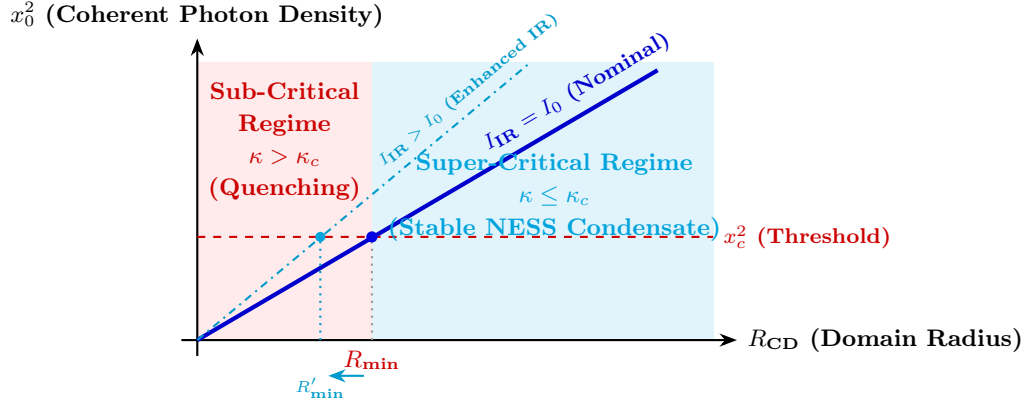


Figure 5: Bifurcation and stability diagram of Coherence Domains as a function of radius R_{CD} . Domains with $R_{CD} < R_{min}$ suffer excessive boundary leakage and quench into thermal chaos. Enhanced IR irradiation ($I_{IR} > I_0$) lowers the critical quenching radius ($R'_{min} < R_{min}$), facilitating macroscopic stabilization.

4. Ionic Debye Screening and Interfacial Electro-Mechanics

Up to this point, we have treated the surrounding fluid as a neutral optical continuum. However, real aqueous, biological, and electrochemical systems are never perfectly pure; they contain significant concentrations of mobile ions, prominently protons (H_3O^+), hydroxide (OH^-), and ubiquitous bulk electrolytes such as sodium (Na^+), potassium (K^+), and chloride (Cl^-).

Rather than acting as disruptive impurities, these mobile charges play a fundamental role in the thermodynamic and mechanical stabilization of the Coherence Domain (CD). In this section, we explain how the ionic plasma interacts with the trapped electromagnetic field to physically “seal” the domain against radiative leakage, and how fluctuations in this ionic shield directly dictate the dynamic stability and mechanical rigidity of the interface.

4.1. Anderson-Higgs Mass Generation and the Ionic Shield

In Section 3, we established that a pristine CD suffers from radiative leakage (κ_{rad}) across its diffuse boundary wall. To understand how real systems overcome this dissipation, we must consider the coupling between the trapped gauge vector potential $\mathbf{A}(\mathbf{r})$ and the mobile ionic currents in the surrounding fluid.

When the evanescent tail of the trapped electromagnetic field penetrates the outer incoherent fluid, it forces the mobile ions into coherent oscillation. According to standard Quantum Field

Theory in continuous media, the interaction between a gauge field and a plasma is governed by the *Anderson-Higgs mechanism* [2, 3]. The mobile ionic cloud acts as a Higgs field, endowing the previously massless propagating photons with an effective mass m_{ph}^* :

$$m_{ph}^* = \frac{\hbar\omega_p}{c^2} \propto \sqrt{\sum_{\alpha} \frac{q_{\alpha}^2 n_{\alpha}}{m_{\alpha}^*}} \quad (25)$$

where ω_p is the local ionic plasma frequency, q_{α} and n_{α} are the charge and number density of the ionic species, and m_{α}^* is their effective solvated mass.

Stabilization by Confinement: The physical consequence of this acquired mass is profound. A “heavy” photon cannot easily propagate through the ionic plasma. Instead of radiating away into infinity, the electromagnetic field is exponentially damped over a characteristic distance corresponding to the Debye screening length, λ_D .

The ionic plasma effectively acts as a conductive mirror. The phenomenological result is that the radiative leakage rate derived in Section 3 is renormalized by an exponential screening factor:

$$\kappa_{eff}(R_{CD}, \lambda_D) \approx \kappa_{rad}(R_{CD}) \exp\left(-\frac{\delta}{\lambda_D}\right) \quad (26)$$

Equation (26) encapsulates a powerful stabilization mechanism: the surrounding electrolytes actively seal the Coherence Domain. By trapping the energy inside, the ionic shield significantly lowers the critical size threshold (R_{min}) required for macroscopic growth.

4.2. Thermodynamic Expulsion and the Double Layer

The local breaking of continuous $U(1)$ gauge symmetry not only traps light but also profoundly alters the local chemical potential of the water molecules. Inside the CD, the phase-locked water molecules occupy a state of lower energy ($\Delta E_{coh} \approx -2.49$ eV) and drastically reduced entropy.

Differential Ion Expulsion: Much like a growing ice crystal expels salt, or a superconductor expels magnetic fields (the Meissner effect), the coherent phase actively excludes non-resonant impurities to maintain its internal phase-locking. Mobile ions—particularly large hydrated cations like Na^+ and K^+ , which are vital to biological systems—are thermodynamically pushed out across the diffuse boundary wall (δ).

Because hydronium ions (H_3O^+) possess a much higher mobility (via the Grotthuss mechanism) and a smaller effective mass than bulky solvated ions (Na^+ , K^+ , Cl^-), they are expelled

more efficiently and organize closest to the interface. This differential expulsion creates a massive spontaneous charge separation:

- The interior of the Coherence Domain becomes effectively electronegative due to the strict exclusion of cations.
- A dense, positively charged proton layer (H_3O^+) forms immediately outside the domain wall.
- A broader, diffuse layer of heavier electrolytes (Na^+ , K^+) forms the outer boundary of this exclusion zone.

This theoretical prediction aligns perfectly with the empirical measurements of Exclusion Zone (EZ) water, which consistently exhibits a negative electrostatic potential of approximately -150 mV relative to the bulk fluid [7].

Interfacial Mechanical Stability: This spontaneous charge separation generates an intense, localized electric field $\mathbf{E}$ across the boundary. To understand the mechanical stability of the domain, we consider the balance of forces acting on the interface. In a stable open system, the domain reaches a mechanical Non-Equilibrium Steady State (NESS) where three macroscopic forces perfectly balance (see Fig. 6):

$$\Delta P_{osm} - P_{Maxwell} - \frac{2\sigma_Q}{R_{CD}} = 0 \quad (27)$$

The physical origin of each term is clear:

- (i) **Osmotic Pressure** ($\Delta P_{osm} > 0$): The dense cloud of excluded ions concentrated just outside the domain wall creates a powerful outward osmotic pressure, pushing against the surrounding fluid.
- (ii) **Dielectrophoretic Maxwell Stress** ($P_{Maxwell} > 0$): The intense electric field interacting with the gradient of dielectric permittivity across the wall exerts an inward electrostatic pull, effectively compressing the domain.
- (iii) **Quantum Surface Tension** (σ_Q): Derived in Section 2, this Laplace-like capillary force resists the expansion of the boundary area.

Equation (27) explains why the CD is not merely a quantum phenomenon, but a robust macroscopic electro-mechanical structure. This equilibrium endows EZ layers with their remarkable structural rigidity, high viscosity, and ability to physically exclude large solutes.

4.3. Thermal Fluctuations and the “Poorly Polished Mirror” Analogy

While Eq. (26) presents the ionic Debye layer as a uniform, static shield (λ_D), any realistic aqueous medium at $T = 300$ K is governed by continuous thermal agitation and hydrodynamic noise. The local ionic concentration is subject to Brownian density fluctuations and local convective micro-currents. Consequently, the local Debye length fluctuates both in space and time: $\lambda_D \rightarrow \lambda_D(\mathbf{r}, t)$.

To develop a physical intuition for the consequences of these fluctuations, we can draw a direct analogy with optical cavity physics.

The Interfacial Plasma as a Rough Mirror: If the ion distribution were perfectly uniform and static, the screening boundary would act as a perfectly smooth plasma mirror, ensuring strict specular reflection and minimizing the leakage rate κ_{eff} .

However, because the ionic density constantly fluctuates, the screening boundary exhibits spatial roughness. The interface acts as a “poorly polished mirror” (or a mirror with a tarnished, uneven silvering) (see Fig. 7). When the trapped coherent mode strikes this rough boundary, the electromagnetic field experiences two distinct loss mechanisms:

1. **Diffuse Scattering:** Spatial irregularities in the ionic index of refraction cause diffuse reflection. Rather than being perfectly reflected inward, a fraction of the trapped mode is scattered outward into the incoherent bulk.
2. **Intermittent Optical Tunneling:** In microscopic regions where thermal fluctuations momentarily deplete the local ion concentration (washing away the peripheral Na^+ and K^+ cloud), the shielding layer thins out (λ_D increases locally). These act as transient “holes” in the mirror, causing sudden bursts of radiative tunneling where trapped photons escape.

Consequences for Dynamic Stability: Because of this poorly polished boundary, the effective leakage rate becomes a fluctuating variable: $\kappa(t) = \bar{\kappa} + \delta\kappa(t)$. This has profound thermodynamic consequences for the stability of the Coherence Domain:

- **Increased Thermodynamic Cost:** To maintain structural integrity, the domain must continuously harvest more ambient IR thermal energy (as per Eq. (16)) to compensate

for the continuous scattering and intermittent bursts of leakage caused by the rough ionic mirror.

- **Hydrodynamic Quenching:** If an external mechanical perturbation (e.g., strong fluid shear, an acoustic shockwave, or a massive depolarization in a biological membrane) severely disrupts the ionic proton layer, the effective mirror is “shattered”. The massive and sudden increase in κ_{eff} overwhelms the IR pumping capacity, driving the trapped field amplitude below the critical threshold. The macroscopic coherent phase is instantly destroyed, relaxing back into an ordinary classical liquid.

This dynamic interplay reveals that Coherence Domains operate as ultra-sensitive biophysical sensors. Their mechanical rigidity and quantum stability are entirely enslaved to the continuous fluctuations of the surrounding Na^+ and K^+ plasma.

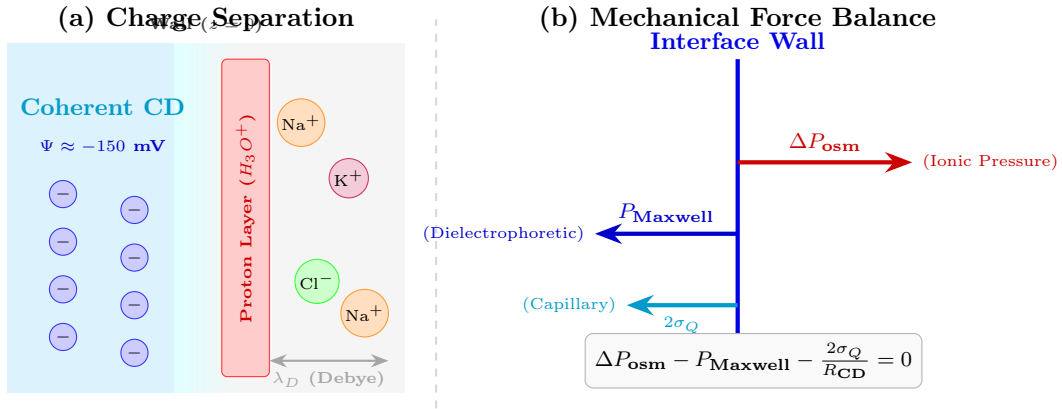


Figure 6: (a) Spontaneous charge separation at the domain boundary: exclusion of cations generates an electronegative core (≈ -150 mV) shielded by a dense protonic shell (H_3O^+) and diffuse Debye plasma (Na^+ , K^+); (b) Mechanical equilibrium of the interface balancing outward osmotic pressure against inward Maxwell stress and quantum surface tension.

5. Surface Anchoring, Bound Water, and the CD–EZ Transition

In bulk liquid water, Coherence Domains (CDs) are continuously subjected to 3D isotropic thermal bombardment and intense radiative leakage, often restricting them to transient, microscopic fluctuations. However, biological systems and microfluidic interfaces are characterized by an abundance of solid surfaces.

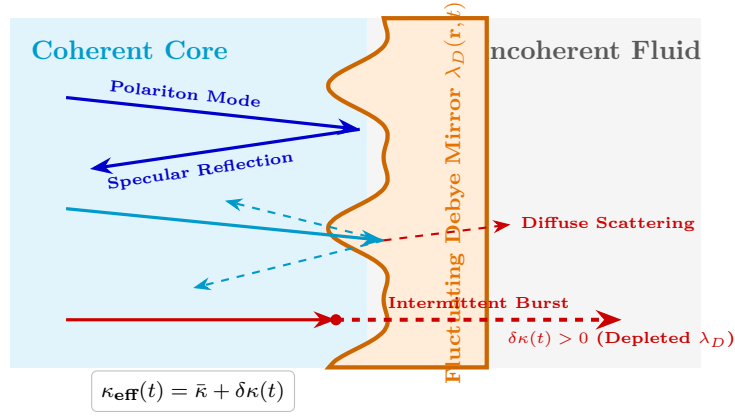


Figure 7: The “poorly polished mirror” analogy: thermal fluctuations in the electrolyte shield generate spatial roughness and local thinning in the Debye screening layer $\lambda_D(\mathbf{r}, t)$. While dense regions ensure specular polariton reflection, local ionic depletion triggers diffuse scattering and intermittent radiative tunneling bursts.

When a CD forms adjacent to a hydrophilic surface, its physical reality fundamentally changes. In this section, we demonstrate how hydrophilic anchoring acts as a structural template that locally reduces the quantum energy threshold, dynamically suppressing boundary leakage, and transforming a microscopic spherical droplet into a massive, highly viscous macroscopic Exclusion Zone (EZ) layer.

5.1. Hydrophilic Boundary Conditions and the “Seeded” Nucleation

A hydrophilic surface (such as a biological membrane, a protein fold, or a Nafion polymer) is characterized by a dense matrix of localized, static electrical charges and hydrogen-bonding sites. When classical, unassociated water molecules approach this surface, their permanent electric dipoles are forcefully aligned by the strong localized electrostatic field of the solid lattice.

To capture this mathematically, we modify the Ginzburg-Landau free energy functional introduced in Eq. (3). The presence of the hydrophilic interface (located at $z = 0$) acts as an external symmetry-breaking field h_S , adding a surface anchoring term:

$$F[\theta(\mathbf{r})] = \int d^3r \left[\frac{1}{2} \rho_{CD} |\nabla \theta(\mathbf{r})|^2 + u(\theta(\mathbf{r})) - u(0) \right] - \int_S h_S \theta(x, y, 0) dx dy \quad (28)$$

The Template Effect: In the bulk fluid, the order parameter must spontaneously break the

vacuum symmetry from $\theta = 0$ to $\theta = \theta_0$ against the full disruptive force of thermal noise, requiring the critical coupling parameter $\gamma > \gamma_c = 1/2$ [1].

However, at the hydrophilic surface ($z = 0$), the external field h_S pre-aligns the molecular dipoles, locking them into a non-zero boundary condition: $\theta(z = 0) \approx \theta_S > 0$. The surface acts as a structural template—or a “seed” in the terminology of crystal nucleation.

Because the boundary molecules are already pre-aligned by the solid, the QED phase transition no longer has to start from scratch. The local critical coupling threshold required to maintain the Mexican Hat potential is dramatically reduced (see Fig. 8):

$$\gamma_{c,eff}(z) \approx \gamma_c \left(1 - \alpha \frac{h_S}{k_B T} e^{-z/\xi} \right) \quad (29)$$

where α is a geometric coupling constant and ξ is the structural correlation length. Equation (29) explains why macroscopic coherence is overwhelmingly favored at hydrophilic interfaces: the solid surface subsidizes the thermodynamic cost of alignment, allowing robust CDs to nucleate and thrive at ambient temperatures where bulk domains might otherwise quench.

5.2. Geometric Shielding and Topological Protection

Beyond lowering the thermodynamic threshold, the solid surface fundamentally alters the dissipative geometry of the domain, rescuing it from the rapid spherical quenching discussed in Section 3.

When a CD anchors to a surface, it unfolds from a microscopic sphere into an extended planar Exclusion Zone (EZ) layer of thickness L_{EZ} . This geometric transition provides immense physical protection:

1. **The Infinite Mirror:** The solid hydrophilic wall ($z \leq 0$) is impenetrable to the trapped vector potential. Radiative leakage is strictly cut off in this half-space. The CD only leaks into the bulk water on its single, outward-facing boundary ($z = L_{EZ}$). As derived in Eq. (22), the geometric leakage rate κ drops drastically by a factor of $R_{CD}/(3L_{EZ})$.
2. **Surface Plasmon Polaritons (SPPs):** At the boundary between the coherent phase (with elevated refractive index $n_{bulk} \approx 1.15$) and the solid substrate, the trapped electromagnetic field can couple to the surface charges, forming Surface Plasmon Polaritons. These topological edge modes guide the coherent energy longitudinally along the 2D

plane (the (x, y) axes in Fig. 1), confining the vector potential tightly to the interface without allowing it to radiate into the 3D bulk.

By combining the “seeded” thermodynamic threshold (Eq. 29) with the halved boundary leakage and topological SPP guiding, the hydrophilic surface transforms the transient CD into a stable macroscopic layer, limited in thickness only by the penetration depth of the ambient IR thermal pump (Section 3).

5.3. Viscosity, Mechanical Rigidity, and the “Quantum Gel”

The macroscopic Exclusion Zone is famous not only for its electrical properties (Section 4) but also for its anomalous rheology. EZ water is highly viscous, mechanically rigid, and exhibits a measurable shear modulus, behaving more like a liquid crystal or a soft gel than ordinary fluid water [7].

Our QED framework provides a rigorous microscopic origin for this “Quantum Gel” behavior.

The “Marching Band” Analogy: In classical bulk water ($\theta = 0$), molecules are held together by transient hydrogen bonds that break and reform on a picosecond timescale. The molecules can freely tumble, rotate, and slide past one another, resulting in the low dynamic viscosity of ordinary water ($\eta_{bulk} \approx 1 \text{ mPa} \cdot \text{s}$). This is analogous to a random crowd of people walking through a plaza; individuals can easily weave through the chaos.

Inside the surface-anchored CD ($\theta \rightarrow \theta_0$), the molecules are not just chemically bonded; their electronic valence clouds are strictly phase-locked to the macroscopic trapped electromagnetic field. They are all oscillating in perfect quadrature. This is analogous to a tightly synchronized marching band.

If a localized shear force attempts to rotate or displace a single water molecule, it must not only break local hydrogen bonds but also break the macroscopic gauge phase-lock. This local phase-slip costs an immense energetic penalty corresponding to the collective gap shift: $\Delta E_{slip} \approx \Delta E_{coh} \sin^2 \theta(z)$.

Macroscopic Rheological Formulation: Using a standard Arrhenius-type transition state model for fluid shear, the effective dynamic viscosity $\eta_{eff}(z)$ inside the coherent layer is exponentially

amplified by the quantum energy gap (see Fig. 9):

$$\eta_{eff}(z) = \eta_{bulk} \exp\left(\frac{\Delta E_{coh} \sin^2 \theta(z)}{k_B T}\right) \quad (30)$$

Because $\Delta E_{coh} \approx 2.49$ eV is nearly two orders of magnitude larger than the thermal energy at room temperature ($k_B T \approx 0.026$ eV), even a partial phase lock ($\sin^2 \theta \ll 1$) results in an exponential explosion of the local viscosity.

Equation (30) dictates that the surface-anchored Coherence Domain physically acts as a rigid, semi-crystalline wall. When combined with the massive outward osmotic pressure exerted by the expelled ionic layer (Eq. 27), this quantum-induced viscosity explains the mechanical resilience of the Exclusion Zone and its ability to physically repel and exclude large colloidal particles and solutes.

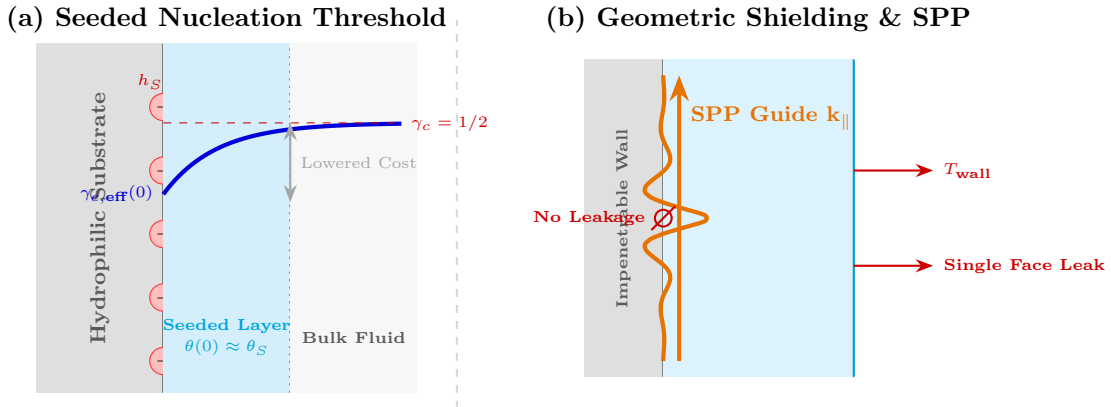


Figure 8: (a) Seeded nucleation at a hydrophilic interface: the external static field h_S locks boundary dipoles into $\theta(0) \approx \theta_S$, drastically lowering the critical threshold $\gamma_{c,eff}(z)$; (b) Planar shielding: the substrate acts as an impenetrable mirror eliminating half-space radiation, while Surface Plasmon Polaritons (SPPs) confine coherent energy along the 2D plane.

6. Physiological and Biophysical Implications

The theoretical framework developed in the preceding sections establishes that Coherence Domains (CDs) and Exclusion Zones (EZ) are not static anomalies, but highly responsive, non-equilibrium electro-mechanical structures. Their macroscopic existence is inextricably linked to continuous infrared pumping (Section 3), ionic Debye screening (Section 4), and hydrophilic surface templating (Section 5).

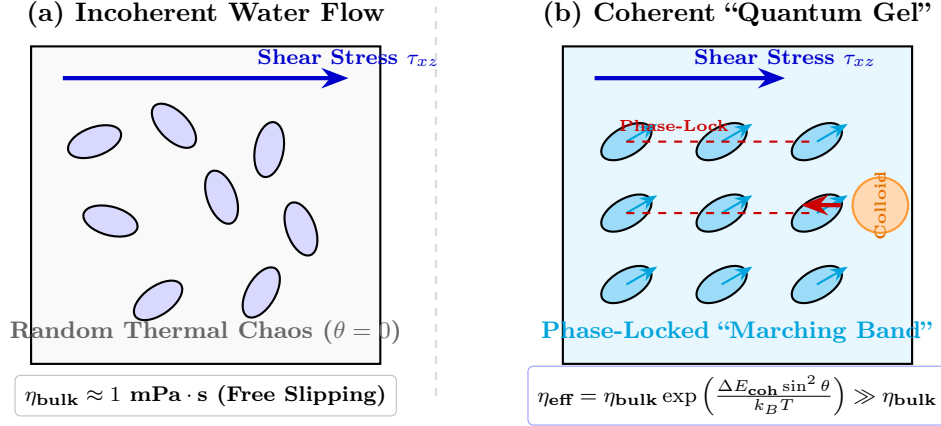


Figure 9: (a) Classical fluid: molecules tumble independently, allowing low-viscosity shear; (b) Coherent EZ layer: collective phase-locking forces molecules into a rigid “marching band”. Breaking local alignment requires a massive phase-slip energy ($\Delta E_{\text{slip}} \approx 2.49 \text{ eV}$), exponentially amplifying viscosity and driving macroscopic solute exclusion.

Because biological systems are inherently composed of dense hydrophilic interfaces (lipid bilayers, folded proteins, nucleic acids) bathed in rich electrolyte solutions (Na^+ , K^+ , Cl^-), they represent the ultimate physical playground for QED water dynamics. In this section, we apply our findings to two fundamental physiological phenomena: the transient decoherence of cell membranes during an action potential, and the non-dissipative transport of protons.

6.1. Membrane Dynamics, Action Potentials, and Transient Decoherence

A resting biological cell membrane is a quintessential hydrophilic template. As demonstrated in Section 5, it “seeds” the nucleation of a massive, highly viscous Exclusion Zone layer. Simultaneously, the active pumping of Na^+ out of the cell and K^+ into the cell establishes a steep ionic gradient. As shown in Section 4, this dense ionic plasma acts as a protective mirror, confining the coherent vector potential and stabilizing the EZ layer.

The resting membrane is therefore shielded by a rigid “Quantum Gel” (Eq. 30) that restricts anomalous molecular diffusion and maintains cellular integrity.

Shattering the Ionic Mirror. During an action potential, voltage-gated ion channels open, triggering a massive, localized influx of Na^+ and efflux of K^+ . From a QED perspective, this chaotic ionic avalanche radically alters the local Debye screening length $\lambda_D(t)$.

The highly ordered ionic double layer that previously shielded the domain is violently disrupted. In the terminology introduced in Section 4.3, the “ionic mirror” is abruptly shattered. The effective boundary leakage rate, which is dynamically coupled to the instantaneous Debye length, spikes dramatically:

$$\kappa_{eff}(t) \approx \kappa_{rad} \exp\left(-\frac{\delta}{\lambda_D(t)}\right) \quad (31)$$

When the ionic structure collapses, $\lambda_D(t)$ effectively increases locally. The exponential shielding factor approaches unity, and $\kappa_{eff}(t) \rightarrow \kappa_{rad}$.

Transient Melting and Symmetry Restoration: As κ_{eff} instantly overwhelms the steady-state thermal IR pumping capacity ($\sigma_{IR}I_{IR}$), the trapped photon amplitude $x(t)$ plummets below the critical quenching threshold ($x < x_c$). The broken $U(1)$ gauge symmetry is transiently restored to the trivial vacuum state.

Macroscopically, this means the coherent phase locally collapses. The phase-lock is broken ($\theta \rightarrow 0$), and the quantum energy gap vanishes ($\Delta E_{coh} \rightarrow 0$). According to our rheological model (Eq. 30), the local viscosity drops exponentially:

$$\eta_{eff}(t) \xrightarrow{\text{depolarization}} \eta_{bulk} \approx 1 \text{ mPa} \cdot \text{s} \quad (32)$$

This transient “melting” of the EZ layer is physiologically crucial. The sudden liquefaction of the interfacial water drastically lowers the hydrodynamic resistance, allowing the massive and rapid flow of ions through the membrane channels required for the action potential to propagate. Once the channels close and the Na^+/K^+ ATPases restore the ionic gradient, the protective ionic mirror reforms, the domain re-nucleates, and the highly viscous EZ gel instantly “refreezes”.

This reveals the action potential not merely as a passive electrical circuit, but as a propagating wave of *quantum phase transition* (melting and refreezing) along the membrane interface.

6.2. Microfluidic Transport and Non-Dissipative Flow along CDs

In addition to membrane dynamics, our framework provides a rigorous explanation for the ultra-fast, seemingly frictionless transport of protons observed in biological microfluidics (such as in ATP synthase or mitochondrial proton wires).

The 2D Protonic Highway: As derived in Section 4.2 (Eq. 27), the thermodynamic expulsion of non-resonant impurities forces hydronium ions (H_3O^+) to accumulate in a dense, highly ordered 2D proton layer immediately outside the CD boundary ($z \approx +\delta$).

Because the interior of the CD acts as a highly rigid, phase-locked “marching band”, it provides a perfectly smooth, topologically protected dielectric substrate. The protons confined to this 2D boundary layer do not experience the chaotic 3D Brownian scattering of the bulk fluid. Instead, they propagate tangentially along the interface via the Grotthuss mechanism (hopping between adjacent hydrogen bonds).

Topological Guiding and Super-Mobility: In the presence of a longitudinal electrochemical gradient $\nabla_{\parallel}\Phi$ (e.g., along a membrane or a microtubule), the protonic current density $\mathbf{J}_H$ is guided along the boundary. Because the adjacent coherent water molecules are locked in a uniform macroscopic phase ($\nabla_{\parallel}\theta \approx 0$), there are no structural defects or thermal fluctuations at the interface to scatter the proton momentum.

The proton mobility μ_H^* along this boundary is therefore drastically enhanced compared to the bulk mobility μ_H^{bulk} , establishing a quasi-superconducting 1D or 2D protonic wire:

$$\mathbf{J}_H = -en_H\mu_H^*\nabla_{\parallel}\Phi, \quad \text{where } \mu_H^* \gg \mu_H^{bulk} \quad (33)$$

Furthermore, as mentioned in Section 5.2, Surface Plasmon Polaritons (SPPs) propagating along this interface can couple with the proton hopping rate, providing directed momentum that rectifies the flow against minor thermal fluctuations. This non-dissipative interfacial transport mechanism explains how biological systems can achieve near-perfect efficiency in energy transduction and signaling across macroscopic distances, bypassing the fundamental limits of classical viscous diffusion.

7. Conclusion

The historical treatment of QED Coherence Domains (CDs) in liquid water has often relied on idealized models of isolated, perfectly sealed cavities. While sufficient to establish the fundamental ground state of Spontaneous Symmetry Breaking, this equilibrium approximation is inadequate for describing the complex, fluctuating reality of open aqueous and biological systems.

In this paper, we have formulated a continuous, dissipative, and spatio-temporal field theory that rigorously bridges the gap between microscopic quantum electrodynamics and macroscopic interfacial biophysics. By incorporating spatial gradients, boundary leakage, thermal pumping, and ionic screening, we have demonstrated that the Coherence Domain is not a static crystalline artifact, but a highly dynamic, Non-Equilibrium Steady State (NESS) dissipative structure.

The primary theoretical paradigm shifts established in this work are:

- **Thermodynamic Inversion (Heat as Fuel):** A bounded CD cannot exist at absolute zero due to inescapable radiative boundary leakage across its diffuse domain wall ($\delta \approx 15$ nm). Ambient thermal infrared radiation at $T = 300$ K is not merely disruptive noise; it provides the vital continuous energy pump required to parametrically sustain the coherent vector potential against dissipation.
- **The Ionic Shield and Mechanical Rigidity:** Solvated ions are not passive impurities. Through the Anderson-Higgs mechanism, the expelled ionic plasma forms a dynamic Debye shield (λ_D) that exponentially suppresses radiative leakage. Furthermore, the thermodynamic expulsion of protons creates a massive spontaneous charge separation, perfectly balancing dielectrophoretic Maxwell stress against ionic osmotic pressure. This electro-mechanical equilibrium is the rigorous microscopic origin of the Exclusion Zone (EZ).
- **Topological Anchoring and the Quantum Gel:** Hydrophilic surfaces act as structural templates that drastically lower the quantum nucleation threshold. By unfolding into planar EZ layers, CDs benefit from halved boundary leakage and topological protection (Surface Plasmon Polaritons). The macroscopic phase-locking of the molecular valence clouds imposes an immense energetic penalty on local shear, explaining the anomalous, gel-like viscosity of interfacial water.

Ultimately, this framework redefines our understanding of biological water. Rather than a passive, neutral solvent, the aqueous medium operates as an active, tunable liquid crystal. Transient fluctuations in the ionic shield dictate the instant melting and refreezing of the coherent phase—driving macroscopic physiological events such as the cellular action potential—while the topologically protected boundary layers provide non-dissipative highways for ultra-fast proton transport.

By integrating quantum field dynamics with classical hydrodynamics and electrochemistry, this model provides a comprehensive, predictive foundation for the next generation of biophysical and microfluidic research.

Appendix A. Derivation of Single-Mode Dynamics from General Multimode Maxwell-Bloch Field Equations

In Section 3.1, the open electro-vibrational dynamics of a Coherence Domain were formulated using a reduced two-variable scalar system for the field amplitude $x(t)$ and vibrational reservoir $y(t)$:

$$\frac{dx}{dt} = (gy - \kappa)x \quad (\text{A.1})$$

$$\frac{dy}{dt} = \sigma_{IR}I_{IR} - gyx^2 \quad (\text{A.2})$$

In this Appendix, we provide the formal derivation showing that Eqs. (A.1)–(A.2) represent the spatio-temporal mean-field, single-mode projection of a much more general *Multimode Open Maxwell-Bloch Field Theory*.

Appendix A.1. The Continuous Multimode Spatial Equations

In a continuous dielectric fluid, the trapped electromagnetic vector potential $\mathbf{A}(\mathbf{r}, t)$ cannot be restricted *a priori* to a single monochromatic wave. In full generality, the radiation field is expanded over a complete orthogonal basis of spatial cavity/polariton modes $\mathbf{u}_m(\mathbf{r})$ with eigenfrequencies ω_m :

$$\mathbf{A}(\mathbf{r}, t) = \sum_m \sqrt{\frac{\hbar}{2\epsilon_0 n_{eff}^2 \omega_m V}} \left[x_m(t) \mathbf{u}_m(\mathbf{r}) e^{-i(\omega_m t - \phi_m(t))} + \text{c.c.} \right] \quad (\text{A.3})$$

where $x_m(t)$ is the dimensionless amplitude of the m -th mode, $\mathbf{u}_m(\mathbf{r})$ satisfies the spatial Helmholtz equation $\nabla^2 \mathbf{u}_m + n_{eff}^2(\mathbf{r}) \frac{\omega_m^2}{c^2} \mathbf{u}_m = 0$ with normalized mode volume $\int_V |\mathbf{u}_m(\mathbf{r})|^2 d^3r = V$, and $m = (\mathbf{k}, \lambda)$ indexes spatial wavevectors and polarization states.

Simultaneously, the intramolecular vibrational excitations are distributed across continuous spatial positions $\mathbf{r}$ and sub-band frequencies ω_μ , forming a spatial excitation density field $y_\mu(\mathbf{r}, t)$.

Coupling the radiation spectrum m to the vibrational reservoir spectrum μ via non-linear optomechanical interactions—and accounting for spatial excitation diffusion and mode-dependent

leakage κ_m —yields the fundamental *Multimode Spatial Rate Equations*:

$$\frac{dx_m(t)}{dt} = \left(\sum_{\mu} g_{m\mu} \int_V \frac{d^3r}{V} |\mathbf{u}_m(\mathbf{r})|^2 y_{\mu}(\mathbf{r}, t) - \kappa_m \right) x_m(t) \quad (\text{A.4})$$

$$\frac{\partial y_{\mu}(\mathbf{r}, t)}{\partial t} = D_{\mu} \nabla^2 y_{\mu}(\mathbf{r}, t) + \sigma_{IR, \mu} I_{IR}(\mathbf{r}, \mu, t) - \gamma_{\mu} y_{\mu}(\mathbf{r}, t) - \sum_m g_{m\mu} |\mathbf{u}_m(\mathbf{r})|^2 y_{\mu}(\mathbf{r}, t) x_m^2(t) \quad (\text{A.5})$$

where:

- $g_{m\mu}$ is the non-linear coupling tensor between electromagnetic mode m and vibrational mode μ .
- $\kappa_m = \frac{v_g}{n_{bulk}} \left(\frac{S}{V} \right)_m T_{wall}(\omega_m)$ is the mode-dependent radiative boundary loss.
- D_{μ} is the spatial diffusion coefficient of vibrational energy in the hydrogen-bond network.
- γ_{μ} is the passive non-radiative vibrational decay rate.
- $\int_V \frac{d^3r}{V} |\mathbf{u}_m(\mathbf{r})|^2 y_{\mu}(\mathbf{r}, t)$ is the spatial overlap integral (governing spatial hole burning).

Appendix A.2. Physical Approximations for Single-Mode Reduction

Equations (A.1)–(A.2) are exact mathematical reductions of the continuous system (A.4)–(A.5) under three fundamental physical approximations:

1. **Single-Mode Monochromatic Condensation (Mode Competition):** Owing to continuous $U(1)$ gauge symmetry breaking and phase-locking, the fundamental spatial mode ($m = 0$ at frequency $\omega_0 = c/\lambda_{CD}$) possesses the lowest boundary loss κ_0 and maximum non-linear gain. Via non-linear mode competition (analogous to laser single-mode selection), this mode reaches oscillation threshold first and saturates the gain, suppressing all higher-order parasitic modes ($x_m \rightarrow 0$ for $m \neq 0$). Thus, the mode sum collapses to $x_0(t) \equiv x(t)$ and $\kappa_0 \equiv \kappa$.
2. **Fast Spatial Diffusion and Reservoir Pooling:** Because intermolecular vibrational diffusion is rapid compared to macro-domain field variations ($D_{\mu} \nabla^2 y_{\mu} \gg \partial_t y_{\mu}$), spatial hole burning is erased, rendering $y_{\mu}(\mathbf{r}, t) \approx y_{\mu}(t)$ spatially uniform. Integrating the

spatial overlap integral over the normalized mode volume yields $\int_V \frac{d^3r}{V} |\mathbf{u}_0(\mathbf{r})|^2 = 1$. Furthermore, pooling the narrow $O - H$ stretching band into a single effective scalar variable $y(t) = \sum_\mu y_\mu(t)$ simplifies the reservoir dynamics.

3. **Dominance of Stimulated Conversion over Passive Decay:** Under ambient thermal pumping ($I_{IR} \approx 100 \text{ W/m}^2$), stimulated coherent conversion (gyx^2) far outpaces intrinsic passive thermal dissipation ($\gamma_\mu y \ll gyx^2$). Neglecting γ_μ leaves non-linear conversion as the dominant sink for vibrational excitations.

Under approximations (1)–(3), the full field equations (A.4)–(A.5) collapse strictly into the mean-field single-mode dynamic equations (A.1)–(A.2) used throughout Section 3.

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